

May 11, 2022

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: NPDES - LLHg
Pace Project No.: 50315017

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on April 28, 2022. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
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CERTIFICATIONS

Project: NPDES - LLHg

Pace Project No.: 50315017

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268

Illinois Accreditation #: 200074

Indiana Drinking Water Laboratory #: C-49-06

Kansas/TNI Certification #: E-10177

Kentucky UST Agency Interest #: 80226

Kentucky WW Laboratory ID #: 98019

Michigan Drinking Water Laboratory #9050

Ohio VAP Certified Laboratory #: CL0065

Oklahoma Laboratory #: 9204

Texas Certification #: T104704355

Wisconsin Laboratory #: 999788130

USDA Soil Permit #: P330-19-00257

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SAMPLE SUMMARY

Project: NPDES - LLHg

Pace Project No.: 50315017

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50315017001	Effluent	Water	04/27/22 14:25	04/28/22 11:15
50315017002	Effluent - duplic	Water	04/27/22 14:28	04/28/22 11:15
50315017003	Field Blank	Water	04/27/22 14:20	04/28/22 11:15
50315017004	Influent	Water	04/27/22 07:00	04/28/22 11:15

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SAMPLE ANALYTE COUNT

Project: NPDES - LLHg

Pace Project No.: 50315017

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50315017001	Effluent	EPA 1631E	WJW	1	PASI-I
50315017002	Effluent - duplic	EPA 1631E	WJW	1	PASI-I
50315017003	Field Blank	EPA 1631E	WJW	1	PASI-I
50315017004	Influent	EPA 245.1	EAE	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

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SUMMARY OF DETECTION

Project: NPDES - LLHg

Pace Project No.: 50315017

Lab Sample ID Method	Client Sample ID Parameters	Result	Units	Report Limit	Analyzed	Qualifiers
50315017001	Effluent					
EPA 1631E	Mercury	0.280J	ng/L	0.50	05/03/22 10:49	
50315017002	Effluent - duplic					
EPA 1631E	Mercury	0.320J	ng/L	0.50	05/03/22 10:57	

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ANALYTICAL RESULTS

Project: NPDES - LLHg

Pace Project No.: 50315017

Sample: Effluent		Lab ID: 50315017001		Collected: 04/27/22 14:25		Received: 04/28/22 11:15		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
1631E Mercury, Low Level		Analytical Method: EPA 1631E Preparation Method: EPA 1631E Pace Analytical Services - Indianapolis							
Mercury	0.280J	ng/L	0.50	0.20	1	05/02/22 16:40	05/03/22 10:49	7439-97-6	

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ANALYTICAL RESULTS

Project: NPDES - LLHg

Pace Project No.: 50315017

Sample: Effluent - duplic		Lab ID: 50315017002		Collected: 04/27/22 14:28		Received: 04/28/22 11:15		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
1631E Mercury, Low Level		Analytical Method: EPA 1631E Preparation Method: EPA 1631E Pace Analytical Services - Indianapolis							
Mercury	0.320J	ng/L	0.50	0.20	1	05/02/22 16:40	05/03/22 10:57	7439-97-6	

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ANALYTICAL RESULTS

Project: NPDES - LLHg

Pace Project No.: 50315017

Sample: Field Blank		Lab ID: 50315017003		Collected: 04/27/22 14:20		Received: 04/28/22 11:15		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
1631E Mercury, Low Level		Analytical Method: EPA 1631E Preparation Method: EPA 1631E Pace Analytical Services - Indianapolis							
Mercury	<0.20	ng/L	0.50	0.20	1	05/02/22 16:40	05/03/22 12:47	7439-97-6	

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ANALYTICAL RESULTS

Project: NPDES - LLHg

Pace Project No.: 50315017

Sample: Influent		Lab ID: 50315017004		Collected: 04/27/22 07:00		Received: 04/28/22 11:15		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
245.1 Mercury Analytical Method: EPA 245.1 Preparation Method: EPA 245.1 Pace Analytical Services - Indianapolis									
Mercury	<0.091	ug/L	0.20	0.091	1	05/10/22 10:25	05/10/22 18:38	7439-97-6	

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: NPDES - LLHg
Pace Project No.: 50315017

QC Batch: 674105 Analysis Method: EPA 1631E
QC Batch Method: EPA 1631E Analysis Description: 1631E Mercury
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50315017001, 50315017002, 50315017003

METHOD BLANK: 3103731 Matrix: Water
Associated Lab Samples: 50315017001, 50315017002, 50315017003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ng/L	<0.20	0.50	0.20	05/03/22 07:56	

METHOD BLANK: 3103732 Matrix: Water
Associated Lab Samples: 50315017001, 50315017002, 50315017003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ng/L	<0.20	0.50	0.20	05/03/22 08:11	

METHOD BLANK: 3103733 Matrix: Water
Associated Lab Samples: 50315017001, 50315017002, 50315017003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ng/L	<0.20	0.50	0.20	05/03/22 11:17	

LABORATORY CONTROL SAMPLE: 3103734

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	ng/L	5	4.80	96	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3103735 3103736

Parameter	Units	92601092001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	ng/L	0.000750 ug/L	3.5	3.5	4.73	4.63	114	111	71-125	2	24	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3103777 3103778

Parameter	Units	50314916002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	ng/L	1.04	4	4	5.20	5.58	104	114	71-125	7	24	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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QUALITY CONTROL DATA

Project: NPDES - LLHg

Pace Project No.: 50315017

QC Batch: 675212

Analysis Method: EPA 245.1

QC Batch Method: EPA 245.1

Analysis Description: 245.1 Mercury

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50315017004

METHOD BLANK: 3109189

Matrix: Water

Associated Lab Samples: 50315017004

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ug/L	<0.091	0.20	0.091	05/10/22 18:29	

LABORATORY CONTROL SAMPLE: 3109190

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	ug/L	5	5.1	102	85-115	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3109191 3109192

Parameter	Units	50315017004 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	ug/L	<0.091	5	5	4.9	4.8	96	96	70-130	0	20	

MATRIX SPIKE SAMPLE: 3109193

Parameter	Units	50315601003 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	ug/L	<0.00020 mg/L	5	5.0	100	70-130	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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QUALIFIERS

Project: NPDES - LLHg

Pace Project No.: 50315017

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: NPDES - LLHg

Pace Project No.: 50315017

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50315017001	Effluent	EPA 1631E	674105	EPA 1631E	674216
50315017002	Effluent - duplic	EPA 1631E	674105	EPA 1631E	674216
50315017003	Field Blank	EPA 1631E	674105	EPA 1631E	674216
50315017004	Influent	EPA 245.1	675212	EPA 245.1	675505

REPORT OF LABORATORY ANALYSIS

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E-CUSTODY / Analytical Request Document

body is a LEGAL DOCUMENT. All relevant fields must be completed accurately.

Section A Required Client Information:		Section C Invoice Information:	
Company: City of Cadillac Address: 200 North Lake St Report To: Cindy Tomaszewski Copy To:	Attention: Cindy Tomaszewski Company Name:	Page: 1 of 1	
Enroll To: ctomaszewski@cadillac-mi.net Phone: 231-775-2368 Requested Due Date/TAT:	Address: Purchase Order No.: Project Name: NPDES - LLHg Project Number:	REGULATORY AGENCY ☒ NPDES ☐ GROUND WATER ☐ DRINKING WATER ☐ UST ☐ RCRA ☐ OTHER	
Site Location		MI	
State Profile #: 710.7		STATE:	

Section D Required Client Information		Valid Matrix Codes		Matrix Code (see valid codes to left)		Sample Type (G=GRAB C=COMP)		COLLECTED				SAMPLE TEMP AT COLLECTION		# OF CONTAINERS		Preservatives										Requested Analysis Filtered (Y/N)		Pace Project No./ Lab I.D.	
ITEM #	SAMPLE ID (A-Z, 0-9 / -) Sample IDs MUST BE UNIQUE	MATRIX	CODE	MATRIX CODE	Sample Type	COMPOSITE START	COMPOSITE END/GRAB	DATE	TIME	DATE	TIME	Unpreserved	H ₂ SO ₄	HNO ₃	HCl	NaOH	Na ₂ S ₂ O ₃	Methanol	Other	Y	N	Residual Chlorine (Y/N)	Pace Project No./ Lab I.D.						
		DRINKING WATER	DW	WATER	WT	WASTE WATER	WW	PRODUCT	P	SOLID	SL		OIL	OL	WIPE	WP	AIR	OT	TISSUE					TS					
1	Effluent			WW	G			4-27	1425							X													
2	Effluent - dupli			WW	G			4-27	1428							X													
3	Field Blank			WT	G			4-27	1420							X													
4	Influent			WW	C	4-26	0700	4-27	0700				X											UNKNOWN					
5																													
6																													
7																													
8																													
9																													
10																													
11																													
12																													
ADDITIONAL COMMENTS				RELINQUISHED BY / AFFILIATION		DATE		TIME		ACCEPTED BY / AFFILIATION		DATE		TIME		SAMPLE CONDITIONS													
low level Hg (mtd. 1631E)				City of Cadillac		4/27/02		1445		Shipped via UPS Ground		4/28/02		1115															

SAMPLER NAME AND SIGNATURE	

PRINT Name of SAMPLER:

SIGNATURE of SAMPLE:

Received on Ice (Y/N)	Custody Sealed Cooler (Y/N)	Samples Intact (Y/N)
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Important Note: By signing this form you are accepting Pace's NET 30 day payment terms and agreeing to late charges of 1.5% per month for any invoices not paid within 30 days.

Sample Conditions Upon Receipt Form (SCUR)

WO#: 50315017
PM: BJH
Due Date: 05/12/22
CLIENT: GR-Cadillac

Date/Time: <u>4/28/22 1144</u>		Evaluated by: <u>AC</u>	
Client: <u>City of Cadillac Uth</u>			
Project Manager: <u>JR/BH</u>		Profile ID:	
Rush TAT Requested: <u>YES</u> <u>NO</u>		Due Date:	
Lab Notified of Rush or Short Holds: <u>YES</u> <u>NO</u>		Non Conformance Form Required: <u>YES</u> <u>NO</u>	
Samples Received Via: FedEx <u>UPS</u> Client Pace Courier Other: _____			Comments:
Custody Seals Present and Intact:	<u>YES</u>	NO	NA
Received Sample Information Form(s): Drinking Waters Only	<u>YES</u>	NO	NA
USDA Regulated Soils: (AL, AR, CA, FL, GA, ID, LA, MS, NM, NY, NC, OK, OR, SC, TN, TX, WA or Puerto Rico)	<u>YES</u>	NO	N/A
Short Holds Present (< 72 Hours):	<u>YES</u>	NO	
Samples Received in Hold:	<u>YES</u>	NO	
Custody Signatures Present:	<u>YES</u>	NO	
Collector Signature Present:	<u>YES</u>	NO	
Packing Material Used:	<u>YES</u>	NO	
Samples Collected Today and On Ice:	<u>YES</u>	NO	N/A
IR Gun #: 280 <u>281</u>	Digital Thermometer #: 282 283		
Ice Type: WET Bagged / WET Loose BLUE <u>NONE</u>	1. Cooler Temp Upon Receipt: <u>15.8/14.8</u> °C		
Ice Location: TOP BOTTOM MIDDLE DISPERSED	Temp should be 0-6°C (Initial/Corrected)		
Temp Blank Received:	<u>YES</u>	NO	
Containers Intact:	<u>YES</u>	NO	
Correct Containers:	<u>YES</u>	NO	
Sufficient Volume:	<u>YES</u>	NO	
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation pH Strip Lot #: <u>H0178689</u> Exceptions are VOA, coliform, LLHg, O&G, or any container with a septum cap or preserved with HCl	<u>YES</u>	NO	N/A
Residual Chlorine Absent: Cl ₂ Strip Lot #: _____ (SVOC/Pest 625, PCB 608, Total/Amenable Cyanide)	<u>YES</u>	NO	N/A
VOA Headspace Acceptable (<6mm):	<u>YES</u>	NO	N/A
Trip Blank Received: HCl MeOH TSP OTHER	<u>YES</u>	NO	
Comments:	2. Cooler Temp Upon Receipt: _____ °C		
	3. Cooler Temp Upon Receipt: _____ °C		
	4. Cooler Temp Upon Receipt: _____ °C		